

Problema 1

% Encontrar FT en lazo cerrado

syms s Kp Kd;

G_s = 10/(s*(s+4));

Gc_s = Kp + Kd*s;

H_s = (Gc_s*G_s)/(1+Gc_s*G_s);

simplifiedExpr = simplify(H_s)

simplifiedExpr =

$$\frac{10 Kp + 10 Kd s}{10 Kp + 4 s + 10 Kd s + s^2}$$

[H_s_num, H_s_den] = numden(simplifiedExpr);

Pc_s = H_s_den

$$Pc_s = 10 Kp + 4 s + 10 Kd s + s^2$$

% Encontrar polinomo deseado a partir de Mp y Ts

Mp = 0.20;

Ts = 1.5;

damping = sqrt(log(Mp)^2/(pi^2 + log(Mp)^2))

damping =

0.4559

Wn = 4/(damping*Ts)

Wn =

5.8486

% encontramos polinomio deseado

Pd_s = s^2 + (2*damping*Wn*s) + Wn^2

Pd_s =

$$s^2 + \frac{16 s}{3} + \frac{4814078648526933}{140737488355328}$$

% Ver numericamente

var = vpa(Pd_s)

$$var = s^2 + 5.3333333333333333333333333333333 s + 34.206086130886127705252874875441$$

% Encontrar constantes

% 1. Extraer los coeficientes de los polinomios con respecto a 's'

% 'coeffs' devuelve un vector [a0, a1, a2] (orden ascendente)

$$G_{c_s} = K_p + (K_i/s) + K_d*s$$

$$G_{c_s} =$$

$$K_p + K_d s + \frac{K_i}{s}$$

$$H_s = (G_{c_s} * G_s) / (1 + G_{c_s} * G_s)$$

$$H_s =$$

$$\frac{4 \left(K_p + K_d s + \frac{K_i}{s} \right)}{\left(\frac{4 \left(K_p + K_d s + \frac{K_i}{s} \right)}{s^2 + \frac{18s}{5} + 9} + 1 \right) \left(s^2 + \frac{18s}{5} + 9 \right)}$$

$$\text{simplifiedExpr} = \text{simplify}(H_s)$$

$$\text{simplifiedExpr} =$$

$$\frac{20 K_d s^2 + 20 K_p s + 20 K_i}{20 K_i + 45 s + 20 K_p s + 20 K_d s^2 + 18 s^2 + 5 s^3}$$

$$[H_{s_num}, H_{s_den}] = \text{numden}(\text{simplifiedExpr});$$

$$P_{c_s} = H_{s_den}$$

$$P_{c_s} = 20 K_i + 45 s + 20 K_p s + 20 K_d s^2 + 18 s^2 + 5 s^3$$

% Encontrar polinomo deseado a partir de Mp y Ts

% Mp = 0.20;

Ts = 3;

% damping = sqrt(log(Mp)^2/(pi^2 + log(Mp)^2))

damping = 0.5

damping =
0.5000

$$W_n = 4 / (\text{damping} * T_s)$$

Wn =
2.6667

% encontramos polinomio deseado

$$P_{d_s} = s^2 + (2 * \text{damping} * W_n * s) + W_n^2$$

$$P_{d_s} =$$

$$s^2 + \frac{8s}{3} + \frac{64}{9}$$

% Ver numericamente

```
var = vpa(Pd_s)
```

```
var =  $s^2 + 2.6666666666666666666666666666667 s + 7.111111111111111111111111111111$ 
```

```
% Orden del polinomio real y deseado son distintos, asignar polos extra
```

```
n_real = polynomialDegree(Pc_s, s)
```

```
n_real =  
3
```

```
n_deseado = polynomialDegree(Pd_s, s)
```

```
n_deseado =  
2
```

```
% Si el sistema es mayor a orden 2, tenemos que agregar polos
```

```
if n_real > n_deseado
```

```
    % Encontrar las raices del polinomio deseado
```

```
    sigma = damping * Wn;
```

```
    p3 = -10*sigma;
```

```
    Pd_s = expand((s - p3)*Pd_s);
```

```
end
```

```
simplify(Pd_s)
```

```
ans =
```

```
 $s^3 + 16 s^2 + \frac{128 s}{3} + \frac{2560}{27}$ 
```

```
% Encontrar constantes
```

```
% 1. Extraer los coeficientes de los polinomios con respecto a 's'
```

```
% 'coeffs' devuelve un vector [a0, a1, a2] (orden ascendente)
```

```
coeffs_c = coeffs(Pc_s, s)
```

```
coeffs_c = (20 Ki 20 Kp + 45 20 Kd + 18 5)
```

```
coeffs_d = coeffs(Pd_s, s)
```

```
coeffs_d =
```

```
 $\left(\frac{2560}{27} \frac{128}{3} 16 1\right)$ 
```

```
% 2. Crear las ecuaciones igualando término a término (término independiente y  
término en s)
```

```
eq1 = coeffs_c(1) == coeffs_d(1);
```

```
eq2 = coeffs_c(2) == coeffs_d(2);
```

```
eq3 = coeffs_c(3) == coeffs_d(3);
```

```
% 3. Resolver el sistema de ecuaciones algebraicas para Kp y Kd
```

```
sol = solve([eq1, eq2, eq3], [Ki, Kp, Kd])
```

```
sol = struct with fields:
```

```
  Ki: 128/27
```

```
  Kp: -7/60
```

```
  Kd: -1/10
```

```
% 4. Convertir los resultados a formato numérico (double)
```

```
Kp_val = double(sol.Kp)
```

```
Kp_val =
```

```
-0.1167
```

```
Kd_val = double(sol.Kd)
```

```
Kd_val =
```

```
-0.1000
```

```
Ki_val = double(sol.Ki)
```

```
Ki_val =
```

```
4.7407
```

```
vpa(subs(Gc_s, [Kp, Kd], [Kp_val, Kd_val]))
```

```
ans =
```

```

$$\frac{Ki}{s} - 0.1 s - 0.11666666666666666666666666666667$$

```